Xavier **Bacon**

Postdoctoral researcher



About me

Postdoctoral researcher

Areas of specialization

Optimal transport
Optimization
Complex systems
Heterogeneity study
Computational aspects
Modeling mathematics

Professional experience

2024-now	**Post-doc** École Normale Supérieure de Paris-Saclay As part of the ANR project SafePaw (Societal Assets For E-healthcare Patient pAthWays project) under the supervision of Julien Randon-Furling.
2020–2023	**A.T.E.R.** Paris Dauphine-PSL University, C.E.R.E.M.A.D.E. team During this period, I developped an optimal resource allocation model that took into account transport costs and implemented it in Python.
2017-2020	**PhD in Applied Mathematics** Panthéon-Sorbonne University, S.A.M.M. team The topic was to explore possible generalizations of optimal transport theory to multivalued densities.

Degrees

2022	**Thesis in Applied Mathematics** Panthéon-Sorbonne University
2017	**Master in *Mathématiques de la Modélisation*** Pierre & Marie Curie University
2016	**Agrégation in Mathematics** Option *Calcul scientifique*
2013	**Bachelor's Degrees in Mathematics and in Physics** Pierre & Marie Curie University

(pre-)Publications

2026	**Optimal transport for primary care planning** (with S. Kane Diallo and J. Touzet) Preprint
2025	**Analysing Spatial Heterogeneity using Optimal Transport** (with J. Touzet) Preprint
2023	**A spatial Pareto exchange economy problem** (with G. Carlier and B. Nazaret) Applied Mathematics and Optimization Journal
2022	**A few models of vector-valued Optimal Transport** Thesis
2020	**Multi-species Optimal Transportation** Journal of Optimization Theory and Applications

Teaching experience (> 600 hours)

Lectures in Linear Algebra, Analysis and Probability (L1) and tutoring in Continuous Optimization (M1), Calculus (L1), Linear Algebra (L2) and Analysis (L2).

Other skills and abilities

Language: Français (native), English (fluent), Japanese (N3 level), German (Basic knowledge).
Programming : Python (Numpy, POT, Pandas, Open-CV, Pynsee), LaTeX, git, godot (on my spare time between my thesis and my post-doc).

Interests

In my spare time, I practice martial arts (Karate 3rd dan, Kudo 5th kyu) and study Japanese language.

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